Gaming the Two Numbers They Control: Quantization, Weight-Cut Bunching, and the Limits of Strength in 3.9 Million Powerlifting Results

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***Abstract*—A powerlifting competitor controls exactly two numbers: the weight loaded on the bar and their bodyweight at weigh-in. We ask whether these choices leave measurable statistical fingerprints, using ∼3.94 million competition results from the public-domain OpenPowerlifting database. Four findings, each answered with a course-appropriate test and led by effect size (at $n \approx 3.9$M almost everything is "significant"). (i) *Quantization*: 96.2% of 13.4M attempt loads sit exactly on the 2.5 kg plate grid; a generalized likelihood-ratio test (GLRT) whose null parameter lies on the boundary is calibrated by a parametric bootstrap. (ii) *Weight-cut bunching*: bodyweight piles up just below every IPF class limit (de-heaped $\log(\text{below}/\text{above}) = +1.92$ at 83 kg); a Cattaneo–Jansson–Ma density-discontinuity test rejects at all seven limits (surviving Holm and Benjamini–Hochberg), a non-limit control shows no bunching, and placebos show no false bunching. (iii) *Allometry*: strength scales with bodyweight differently by sex ($b = 0.75$ men, 0.51 women vs. the isometric $2/3$; a Sex$\times \log$(BW) interaction is highly significant and persists on common support). (iv) *Prediction*: basic controllable variables predict the total ($R^2 = 0.70$, random forest, leakage-safe grouped CV). The strategy leaves a measurable fingerprint. Code and data: https://github.com/adirelm/opl-stat-theory.**

I. Introduction

Powerlifting comprises three lifts (squat, bench press, deadlift), three attempts each, contested within bodyweight classes. A competitor's agency reduces to two numbers: the load chosen for each attempt, and the bodyweight registered at weigh-in. We treat these as a natural laboratory for statistical signatures of strategic behaviour, and ask what ∼3.94M results reveal about the limits of human strength.

Related work. Weight cutting is well documented in weight-class sports, and bunching/manipulation around administrative thresholds is classically tested with the McCrary density test [1] and its modern local-polynomial form [2]. Allometric (square–cube) scaling predicts strength $\propto$ bodyweight$^{2/3}$. Extreme-value theory has been used to estimate ultimate athletic records [5]. On OpenPowerlifting specifically, recent work studies *scoring/scaling* [6]; to our knowledge no formal manipulation test of weight-cut bunching has been applied to this dataset.

Contribution. We tell one story—"the two numbers and the limits of strength"—spanning classical inference and machine learning: (1) a rounded-likelihood *quantization* GLRT with boundary-bootstrap calibration; (2) the first formal density-discontinuity test of weight-cut *bunching* on OpenPowerlifting, with a non-limit control, placebos, de-heaping and multiple-testing corrections; (3) a sex-interaction *allometry* model; and

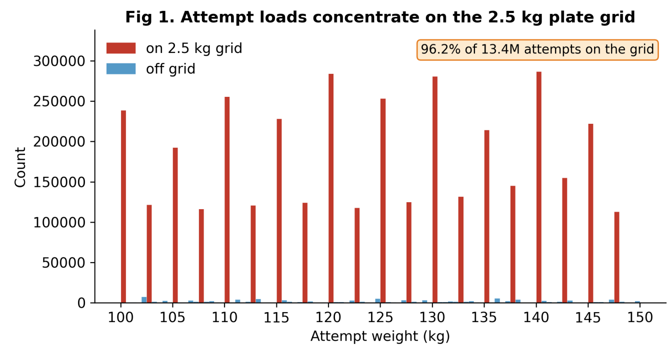

Fig. 1. H1: attempt loads concentrate on the 2.5 kg plate grid.

(4) a leakage-safe strength-*prediction* model. Throughout we lead with effect sizes and confidence intervals rather than p-values.

II. Results

All intervals are 95%. Headline effect sizes use all rows; formal tests use one row per lifter (to respect independence) and, where the class scheme matters, the modern kg-only era (2014+).

A. Attempt loads are quantized to the 2.5 kg grid (H1)

Of 13,397,702 valid attempt loads, **96.20%** (CI $[96.19, 96.21]$) fall exactly on the 2.5 kg plate grid (Fig. 1); the smallest standard plate is 1.25 kg, loaded on both sides. We model the within-cell position on the kg-native (0.5 kg) lattice as a spike-plus-uniform mixture and test the spike weight $\pi = 0$ versus $\pi > 0$ with a GLRT. Because $\pi = 0$ lies on the boundary of the parameter space, Wilks' χ^2 is invalid; the null is a 50:50 mixture of χ^2_0 and χ^2_1 [3], [4], which our parametric bootstrap recovers (mass at zero $=$ 0.51; calibrated 95% cutoff 2.59 vs. the naive 3.84). The observed statistic ($2 \ln \Lambda = 3.9 \times 10^7$) exceeds every bootstrap draw ($p < 3.3 \times 10^{-4}$). The 3.8% off-grid loads are largely a units artifact: of the 2.1% that fall off the 0.5 kg lattice entirely, 82% are round pound loads (e.g., 102.06 kg $=$ 225 lb), not noise.

B. Bodyweight bunches below class limits (H2)

Bodyweight piles up just below real class limits (Fig. 2). On de-heaped data (round/half-kg weigh-ins removed), the log-

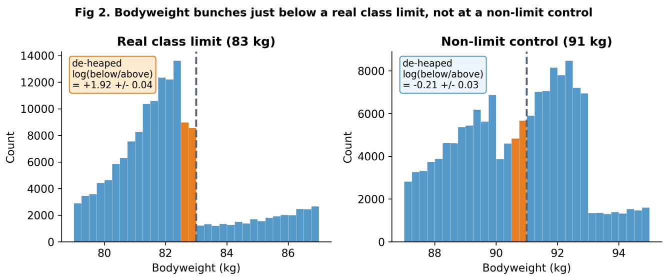

Fig. 2. H2: bunching just below a real limit (83 kg) but not at a non-limit control (91 kg); round-number weigh-ins removed.

TABLE I
DE-HEAPED log(BELOW/ABOVE) AT EACH IPF MEN'S LIMIT (ALL ROWS). GREY ROW IS A NON-LIMIT CONTROL.

Limit (kg)	log-ratio	$\pm$	$\times$ asymmetry
59	$+0.99$	0.04	2.7
66	$+1.18$	0.04	3.2
74	$+1.00$	0.03	2.7
83	$\mathbf{+1.92}$	0.04	**6.8**
93	$+1.65$	0.04	5.2
105	$+1.17$	0.04	3.2
120	$+0.74$	0.06	2.1
91 (control)	-0.21	0.03	0.8

ratio of just-below to just-above counts is positive at all seven IPF men's limits and strongest at 83 kg ($+1.92 \pm 0.04$, a $6.8\times$ asymmetry; Table I, Fig. 3); a non-limit control at 91 kg is flat/slightly negative (-0.21). The formal Cattaneo–Jansson–Ma density-discontinuity test [2] (one random meet per lifter, 2014+) rejects at all seven limits with the density dropping at the cutoff ($t < 0$), surviving both Holm and Benjamini–Hochberg correction; the control shows no excess below (a small *opposite*-direction discontinuity) and no placebo shows the bunching pattern. The effect is robust across equipment (raw $+2.14$, equipped $+1.51$) and persists on the clean kg-only subset ($+1.75$).

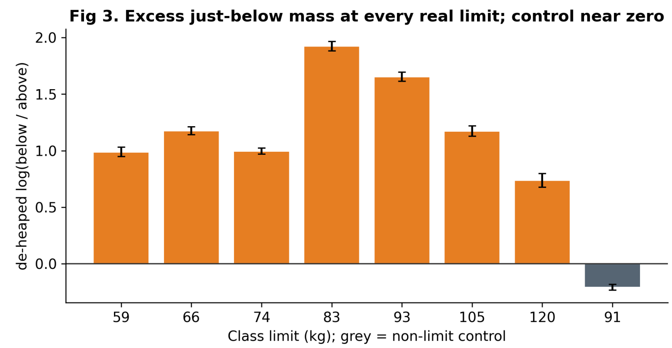

Fig. 3. H2: excess just-below mass at every real limit; control near zero.

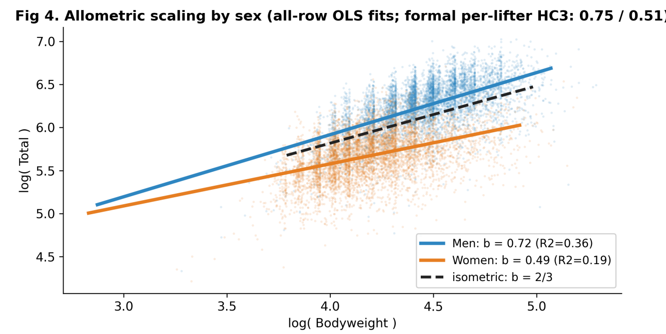

Fig. 4. H3: allometric scaling by sex vs. the isometric 2/3.

C. Allometric scaling differs by sex (H3)

Fitting $\log(\text{Total}) = a + b\log(\text{BW})$ (Fig. 4), the exponent is $b = 0.75$ (CI $[0.74, 0.75]$) for men and 0.51 ($[0.50, 0.51]$) for women, using heteroskedasticity-robust (HC3) standard errors on one row per lifter. A Wald test rejects the isometric $b = 2/3$ for both sexes; framed one-sided, the framing is motivated by Neyman–Pearson/UMP logic (exact UMP only under a one-parameter exponential-family model). A pooled Sex$\times\log(\text{BW})$ interaction estimates the men-minus-women slope gap at $+0.24$ ($p \ll 10^{-3}$), and the gap *grows* on the common bodyweight support (0.84 vs. 0.38), so it is not an artifact of the sexes occupying different weight ranges. The message is not that women are weaker, but that the estimated scaling differs.

D. Strength is predictable from basic variables (H4)

Predicting the total from controllable/basic variables (bodyweight, sex, equipment, age), with every split grouped by lifter and fold-local preprocessing, a random forest reaches $R^2 = 0.70$ (RMSE 86 kg) versus 0.61 for linear regression (Fig. 5); bodyweight and sex dominate the permutation importance. A logistic "made-weight" classifier built from *non-bodyweight* features only (age, tested status, era, equipment; Dots excluded as it is a function of total and bodyweight) reaches AUC $= 0.58$—honestly weak, since cutting is driven by bodyweight relative to the limit, which the label encodes and the features deliberately exclude.

E. Supporting analyses

The apparent two-component structure of raw totals is just sex: by BIC and a bootstrap likelihood-ratio test, raw Total prefers two Gaussian components but the sex-normalized Dots score prefers one (Fig. 6). Untested competitions show higher Dots than tested ones (medians 344 vs. 318, rank-biserial -0.20). An extreme-value (GEV) fit to equal-size block maxima of recent totals gives shape $\xi = -0.02$ with CI $[-0.20, 3.77]$: a population strength ceiling is *not* supported by the data (the CI includes zero). Median Dots has drifted down over 2000–2024 (Spearman $\rho = -0.54$), consistent with broadening participation. Finally, after deduplicating to one row per lifter, the association between tested status and cutting is negligible (Cramér's $V = 0.008$), illustrating how pseudo-replication inflates significance.

III. METHODS

Data. A pinned snapshot of OpenPowerlifting (3,941,811 rows, public domain; SHA-256 recorded for reproducibility).

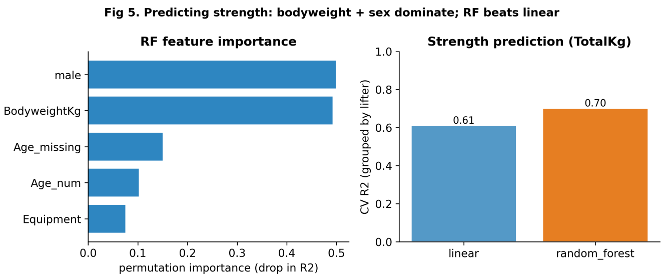

Fig. 5. H4: strength prediction—importance and grouped-CV R^2.

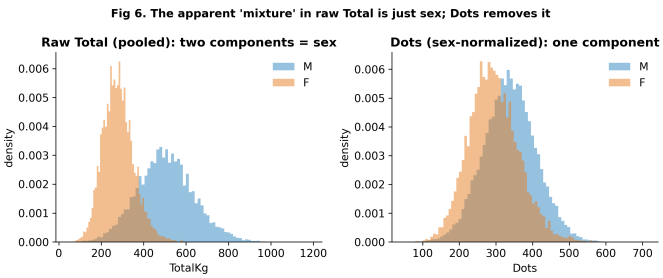

Fig. 6. The apparent "mixture" in raw Total is sex; Dots removes it.

The unit of analysis is the *attempt* for H1 (the mechanism is within-competitor) and one *row per lifter* for the cross-sectional formal tests; de-heaping removes round/half-kg weigh-ins. Because $n \approx 3.9$M makes almost any departure significant, we report effect sizes with CIs and apply multiple-testing corrections (Bonferroni, Holm, Šidák, Benjamini–Hochberg) to pre-declared families.

Tests. H1 uses MLE and a GLRT ($2 \ln \Lambda \sim \chi^2$ under regularity); at the boundary we calibrate the null by parametric bootstrap [3], [4] and report a χ^2 goodness-of-fit. H2 uses the Cattaneo–Jansson–Ma local-polynomial density-discontinuity (manipulation) test [2], with a non-limit control, placebos and de-heaping. H3 uses OLS in log-log space with HC3 and clustered SEs, a Wald test against $2/3$, and a Sex$\times \log$(BW) interaction; correlations report Fisher-transform CIs. H4 uses linear and random-forest regression and logistic classification with grouped-by-lifter cross-validation (R^2/Adjusted-R^2/RMSE/MAE, permutation importance, VIF, AUC). Supporting analyses add a bootstrap mixture-LRT (AIC/BIC), a GEV extreme-value fit, a Wald sequential probability ratio test (with stopping time and expected sample size), one- and two-sample nonparametric tests (Wilcoxon signed-rank, Mann–Whitney), an F-test/Kruskal–Wallis with post-hoc comparisons, and a χ^2 independence test with Cramér's V and standardized residuals.

IV. Discussion

Conclusions. The two numbers a competitor controls leave measurable, falsification-surviving traces: attempt loads are quantized to the plate grid, and bodyweight bunches specifically below class limits, and not at controls or placebos in the bunching direction. Strength scaling is sex-specific, and basic controllable variables already explain $\sim$70% of the variance in the total.

Limitations and null results. This is observational: we identify population-level patterns consistent with weight cutting, not individual intent. The 120 kg cutoff—adjacent to the unbounded 120+ class—shows the weakest bunching, consistent with weaker incentives. The EVT analysis does *not* support a strength ceiling (wide ξ CI), and we find no material association between tested status and cutting; we report these null results as findings. Lifter identity is name-based, the per-lifter meet is chosen at random (we report sensitivity), and the formal bunching test is restricted to the modern kg-only era. These are the natural next steps for the full study.

References

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